

Supporting Information for “A \$100 Arduino-Based Stability Measurement System for Perovskite Solar Cells”

Andrew Chen Co-Authors

CONTENTS

I	Associated Links	2
I-A	Gerber and 3D Enclosure Files	2
II	Detailed Component Specifications	2
II-A	PCB schematics and layout diagrams	2
II-B	Arduino microcontroller specifications	3
II-C	DAC and ADC component datasheets and configurations	4
II-D	Detailed parts list with supplier information and cost breakdown	4
II-E	3D-Printed Components and Sample Holder	4
II-F	Power and Inter-module Communication	4
II-G	Illumination System Specifications	4
III	Assembly Instructions and Guidelines	4
III-A	Step-by-step PCB assembly guide	4
III-B	Instructions for soldering and wiring	4
III-C	3D printing guidelines, files, and recommended print settings	4
III-D	Enclosure assembly and sample holder setup	4
III-E	Troubleshooting common assembly issues	4
IV	Software Implementation and Usage	4
V	Additional GUI Screenshots and Features	4
V-A	Python control software setup instructions	4
V-B	Installation and dependency management (e.g., PyQt, serial communication)	4
V-C	GUI user manual (detailed screenshots and explanations)	4
V-D	Example scripts for automated testing procedures	4
VI	Additional Experimental Data	4
VI-A	Raw JV curves from diode baseline validation	4
VI-B	Complete dataset for separate holder protocol experiment	4
VI-C	Complete dataset for same holder protocol experiment	4
VI-D	Raw data and extended analysis for repeatability measurements	4
VI-E	Extended dataset and degradation curves from 200-hour stability test	4
VII	Future Development and Community Contributions	4
VII-A	Roadmap for integration with environmental chambers and advanced sensing components	4
VII-B	Guidelines for community-driven enhancements and custom modifications	4
VII-C	Instructions for contributing improvements and reporting issues on the open-source repository	4

I. ASSOCIATED LINKS

For convenience, we provide direct links to the design files, and CAD models referenced throughout this work.

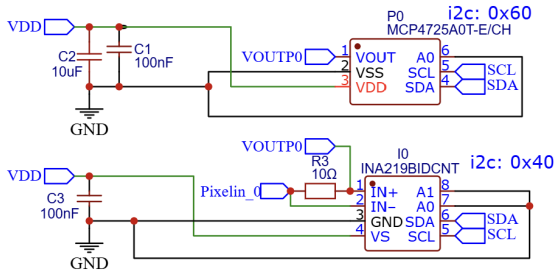
- SMS Main PCB [OSHWLab Link]
- Holder PCB [OSHWLab Link]
- 3-D Printed Enclosure [OnShape Link]

A. Gerber and 3D Enclosure Files

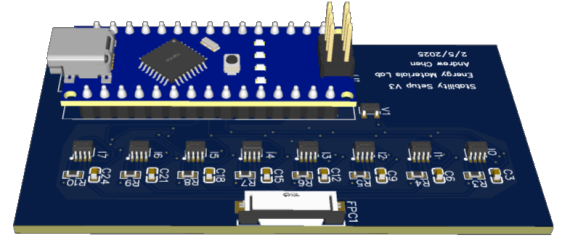
All Gerber files, drill files, and STL models are provided in the GitHub repository: <https://github.com/yourrepo/SMS>.

II. DETAILED COMPONENT SPECIFICATIONS

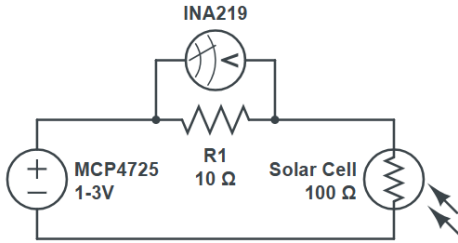
A. PCB schematics and layout diagrams



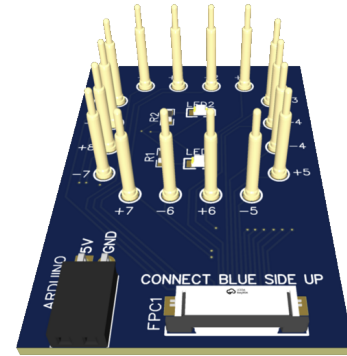
(a) Detailed schematic for a single measurement channel.



(b) Rendered 3D view of the main PCB layout.



(c) Simplified circuit diagram for a single cell assembly.



(d) Assembled measurement probe with pogo pins.

Fig. 1: Electrical diagrams and component layouts of the stability measurement system. (a) Detailed schematic of a single measurement channel, showing the DAC, ADC, and shunt resistor configuration. (b) 3D rendering of the assembled main PCB. (c) Simplified circuit diagram illustrating the basic measurement principle. (d) Photograph of the assembled probe head.

The main board of the Stability Measurement System (SMS) was designed in EasyEDA, with fabrication and assembly handled by JLCPCB. All design files, including schematics and PCB layouts, are open-sourced to facilitate replication and modification by other researchers. The board is engineered to test up to eight solar cell pixels in parallel, with each pixel monitored by an independent sub-circuit.

The core of each sub-circuit, detailed in the schematic in Figure 1a, consists of two primary integrated circuits (ICs) that operate over an I²C bus:

- **MCP4725 DAC:** A digital-to-analog converter used to apply a precise load voltage to the solar cell pixel. Each of the eight DACs is assigned a unique I²C address from 0x60 to 0x67. It can supply voltage from 0-3.3V with a 0.8 mV resolution.
- **INA219 ADC:** A high-side current and voltage sensor that monitors the cell's output. Each of the eight ADCs is assigned a unique I²C address from 0x40 to 0x47. The output is connected to a 10 Ω shunt resistor which is measured across to find voltage and current. This gives a voltage measurement range of 0 to 32 mA with a 8 μ A resolution and a voltage range of 0 to 27 V with a 4 mV resolution.

In the measurement circuit (Figure 1a), the MCP4725 DAC (P0) applies a variable load voltage, ranging from 0 to 3.3 V with a 12-bit resolution (approximately 0.8 mV per step), to the connected solar cell pixel. The INA219 ADC (I0) then measures the resulting voltage and current. Current is measured by monitoring the voltage drop across a high-precision 10 Ω shunt resistor (R3). This configuration allows for current measurements in the range of 0-32 mA with a resolution of 8 μ A. To ensure stable power delivery and reduce signal noise, decoupling capacitors (C1, C2, C3) are placed on the voltage input line.

The entire measurement process for all eight channels is orchestrated by an Arduino Nano (AN), which communicates with the DAC and ADC components via the I²C protocol. The selection of the MCP4725 and INA219 components was based on their low cost, high availability, and robust software support, with well-documented C++ libraries provided by Adafruit. The measurement ranges afforded by these components are well-suited for typical perovskite solar cell characterization. For applications requiring higher precision or different voltage/current ranges, these components can be substituted with pin-compatible alternatives. Figure 1b shows a 3D rendering of the final PCB layout, while Figure 1d shows the assembled holder PCB used for contacting the solar cell pixels.

B. Arduino microcontroller specifications

The system is controlled by an Arduino Nano, selected for its compact form factor, widespread availability, and extensive community support. The Nano is based on the ATmega328P microcontroller, which provides sufficient computational power for managing the measurement tasks. The key technical specifications are outlined in Table I.

TABLE I: Arduino Nano Technical Specifications.

Specification	Value
Microcontroller	ATmega328P
Operating Voltage	5V
Clock Speed	16 MHz
Flash Memory	32 KB (2 KB used by bootloader)
SRAM	2 KB
EEPROM	1 KB
Digital I/O Pins	14
Analog Input Pins	8
I ² C Pins	A4 (SDA), A5 (SCL)

A critical parameter for this application is the microcontroller's flash memory, which dictates the maximum program size. The firmware for the SMS, which includes libraries for the DAC and ADC components as well as the control logic for running the stability tests, occupies approximately 14 KB. This leaves ample space within the Arduino Nano's 32 KB of flash memory for future expansion or the addition of more complex measurement routines.

While the system was designed around the Arduino Nano, it is compatible with other Arduino boards. The primary requirements for a substitute board are sufficient flash memory to accommodate the firmware and support for the I²C communication protocol, which requires at least one set of SDA and SCL pins. Boards such as the Arduino Uno or Arduino Mega meet these criteria and could be used with minimal modification to the firmware's pin definitions.

TABLE II: Bill of materials and cost breakdown for five Stability Measurement System (SMS) modules. PCB prices include fabrication, component sourcing, and assembly. Users may source parts or assemble by hand, though SMD soldering is required.

Component	Qty	Unit Cost (\$)	Total Cost (\$)
Arduino Nano (non-genuine OK)	5	7.00	35.00
Main PCB assembly	5	40.50	202.50
Holder PCB	5	11.40	57.00
3D-printed enclosure (Black PLA)	5	2.00	10.00
LED grow light	1	60.00	60.00
Cables & connectors	10	2.00	20.00
Grand Total			384.50

C. DAC and ADC component datasheets and configurations

D. Detailed parts list with supplier information and cost breakdown

E. 3D-Printed Components and Sample Holder

F. Power and Inter-module Communication

G. Illumination System Specifications

III. ASSEMBLY INSTRUCTIONS AND GUIDELINES

A. Step-by-step PCB assembly guide

B. Instructions for soldering and wiring

C. 3D printing guidelines, files, and recommended print settings

D. Enclosure assembly and sample holder setup

E. Troubleshooting common assembly issues

IV. SOFTWARE IMPLEMENTATION AND USAGE

V. ADDITIONAL GUI SCREENSHOTS AND FEATURES

Figure is the Stability Setup GUI. The Python-based GUI, built with PySide6, is designed to facilitate use of the Stability Setup Hardware. This intuitive interface simplifies system operation and highlights the performance metrics that are critical for evaluating solar panel stability. Here are the functions of the GUI:

- 1) **Set Up Tests:** Enter details like trial names, email notifications, measurement parameters, etc.
- 2) **Manage Settings:** Save presets to easily switch between best configurations.
- 3) **Visualize Data:** Built in live plotting of JV and MPPT curves.
- 4) **Manage Hardware:** Check how many stability setup systems are connected and configure their IDs.

A. Python control software setup instructions

B. Installation and dependency management (e.g., PyQt, serial communication)

C. GUI user manual (detailed screenshots and explanations)

D. Example scripts for automated testing procedures

VI. ADDITIONAL EXPERIMENTAL DATA

A. Raw JV curves from diode baseline validation

B. Complete dataset for separate holder protocol experiment

C. Complete dataset for same holder protocol experiment

D. Raw data and extended analysis for repeatability measurements

E. Extended dataset and degradation curves from 200-hour stability test

VII. FUTURE DEVELOPMENT AND COMMUNITY CONTRIBUTIONS

A. Roadmap for integration with environmental chambers and advanced sensing components

B. Guidelines for community-driven enhancements and custom modifications

C. Instructions for contributing improvements and reporting issues on the open-source repository